

WORKED EXAMPLE

XOR asks whether the inputs differ

Exclusive OR returns 1 for exactly one active input.

$$(x, y) = (0, 0) \Rightarrow 0, \quad (0, 1) \Rightarrow 1$$

$$(x, y) = (1, 0) \Rightarrow 1, \quad (1, 1) \Rightarrow 0$$

The two positive points lie at opposite corners of the unit square. No single straight line separates them from both negative corners.

Lecture 2, page 56: one threshold neuron cannot compute XOR. Hidden neurons create intermediate Boolean features that the output can combine.

$$H(s) = 1 \text{ if } s \geq 0; \quad H(s) = 0 \text{ if } s < 0$$

Keep this inclusive equality convention for every neuron in the lesson.

WORKED EXAMPLE

The three-neuron construction

These are exactly the two hidden gates and output gate on Lecture 2, page 57.

$$h_{\text{OR}} = H(x + y - 1)$$

$$h_{\text{NAND}} = H(-x - y + 1)$$

$$q = H(h_{\text{OR}} + h_{\text{NAND}} - 2)$$

The hidden OR says at least one input is 1. Hidden NAND says they are not both 1. The output AND requires both statements to be true.

NAND has weights (-1, -1), threshold $T = -1$ and bias +1. OR has threshold 1; the output has threshold 2.

Compute hidden outputs first. Feed their 0/1 outputs, not their scores, into the final neuron.

WORKED EXAMPLE

Three neurons: work through (0, 0), (0, 1)

First the two hidden decisions, then the output decision.

$$(x, y) = (0, 0): \quad h_{\text{OR}} = H(0 + 0 - 1) = H(-1) = 0$$

$$h_{\text{NAND}} = H(-0 - 0 + 1) = H(1) = 1$$

$$q = H(0 + 1 - 2) = H(-1) = 0$$

$$(x, y) = (0, 1): \quad h_{\text{OR}} = H(0 + 1 - 1) = H(0) = 1$$

$$h_{\text{NAND}} = H(-0 - 1 + 1) = H(0) = 1$$

$$q = H(1 + 1 - 2) = H(0) = 1$$

WORKED EXAMPLE

Three neurons: work through (1, 0), (1, 1)

First the two hidden decisions, then the output decision.

$$(x, y) = (1, 0): \quad h_{\text{OR}} = H(1 + 0 - 1) = H(0) = 1$$

$$h_{\text{NAND}} = H(-1 - 0 + 1) = H(0) = 1$$

$$q = H(1 + 1 - 2) = H(0) = 1$$

$$(x, y) = (1, 1): \quad h_{\text{OR}} = H(1 + 1 - 1) = H(1) = 1$$

$$h_{\text{NAND}} = H(-1 - 1 + 1) = H(-1) = 0$$

$$q = H(1 + 0 - 2) = H(-1) = 0$$

WORKED EXAMPLE

The two-neuron construction

Lecture 2, page 58 allows direct input-to-output connections.

$$h = H(x + y - 2), \quad q = H(x + y - 2h - 1)$$

The hidden neuron h is AND. The output receives x , y and h , with weights 1, 1 and -2. Its threshold is 1.

If both inputs are 1, the hidden neuron subtracts 2 at the output. This suppresses the 11 case that a simple OR would accept.

The direct edges bypass the hidden neuron. This is why this construction can use fewer neurons than the previous adjacent-layer construction.

Both constructions compute the same XOR truth table. Their allowed connections are different.

WORKED EXAMPLE

Two neurons: work through (0, 0), (0, 1)

Use the newly calculated hidden output in the output score.

$$(x, y) = (0, 0): \quad h = H(0 + 0 - 2) = H(-2) = 0$$

$$q = H(0 + 0 - 2(0) - 1) = H(-1) = 0$$

$$(x, y) = (0, 1): \quad h = H(0 + 1 - 2) = H(-1) = 0$$

$$q = H(0 + 1 - 2(0) - 1) = H(0) = 1$$

The raw inputs go directly to the output as well as to h. Never replace x and y by h.

WORKED EXAMPLE

Two neurons: work through (1, 0), (1, 1)

Use the newly calculated hidden output in the output score.

$$(x, y) = (1, 0): \quad h = H(1 + 0 - 2) = H(-1) = 0$$

$$q = H(1 + 0 - 2(0) - 1) = H(0) = 1$$

$$(x, y) = (1, 1): \quad h = H(1 + 1 - 2) = H(0) = 1$$

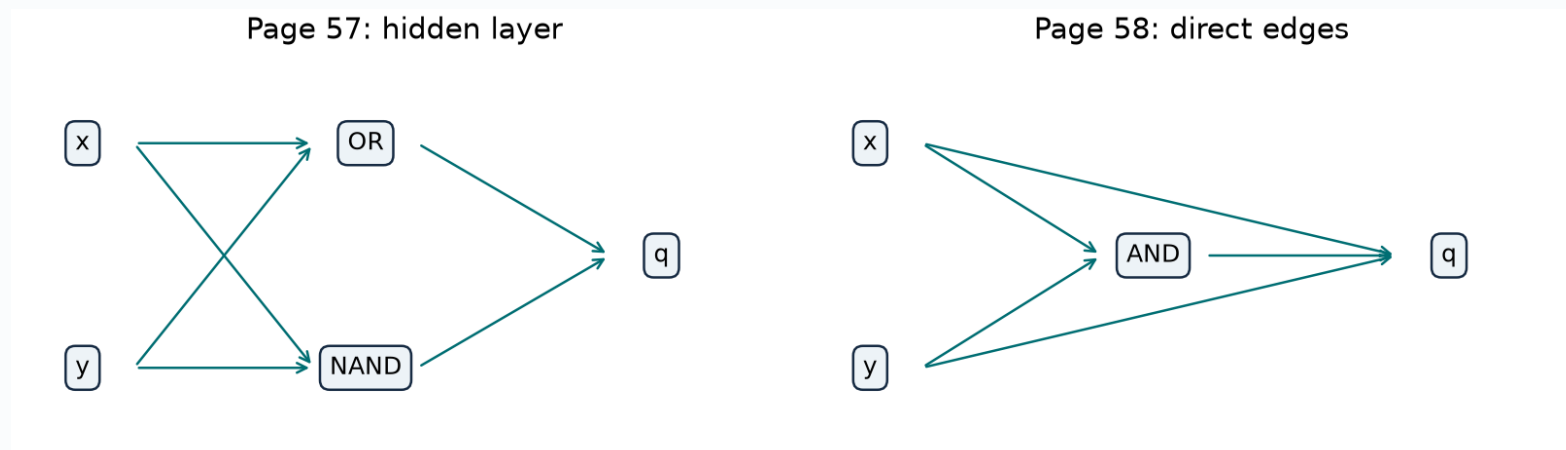
$$q = H(1 + 1 - 2(1) - 1) = H(-1) = 0$$

The raw inputs go directly to the output as well as to h. Never replace x and y by h.

WORKED EXAMPLE

Compare the actual connection patterns

Counts exclude input nodes. Every neuron contributes one bias parameter.



Left: 2 hidden + 1 output = 3 neurons; 4 input edges + 2 output edges = 6 weights; 6 + 3 biases = 9 parameters.

Right: 1 hidden + 1 output = 2 neurons; 2 hidden-input + 3 output-input edges = 5 weights; 5 + 2 biases = 7 parameters.

Depth is 2 in both: the longest path from input to output crosses two computational neurons. These are construction-specific counts.

INDEPENDENT PRACTICE

Your turn: equal inputs should return 1

XNOR is the complement of XOR. Add a NOT neuron after the XOR output q .

$$r = H(-q)$$

1. Use the XOR outputs for 00, 01, 10 and 11. Calculate the new score $-q$ and output r in each row.
2. How many neurons, weights and biases did this added gate introduce?
3. What is the new longest input-to-output depth?

The new gate uses weight -1 and bias 0 . Count its bias as a parameter even though its chosen value is zero.

WORKED ANSWER

Practice answer: one more layer

q is XOR; r is XNOR, so r must be 1 precisely when the original inputs agree.

$$00 : q = 0 \Rightarrow r = H(-0) = H(0) = 1$$

$$01 : q = 1 \Rightarrow r = H(-1) = 0$$

$$10 : q = 1 \Rightarrow r = H(-1) = 0$$

$$11 : q = 0 \Rightarrow r = H(-0) = H(0) = 1$$

Added: 1 neuron, 1 weight and 1 bias. Longest depth rises from 2 to 3 because every old path now ends at the extra NOT neuron.

Avoid: treating a circle threshold as a bias; forgetting direct edges in the two-neuron version; sending hidden scores instead of hidden outputs.